

Renesas's AI accelerator, DRP-AI, combines a dynamic reconfigurable processor (DRP) and AI-MAC, offering energy efficiency that negates the need for cooling mechanisms like fans or heat sinks, making AI integration cost-effective across various sectors, from consumer electronics



to POS terminals. The system-on-chip (SoC) showcases dual Cortex-A55 cores, clocking up to 1GHz, paired with the Cortex-M33, and is enriched by a 3D graphics engine and an H.264 video codec. These components can function within temperature ranges of 0°C to +70°C for commercial settings and an extended -40°C to +85°C for industrial scenarios.

ARIES Embedded

<https://www.aries-embedded.com>

New solution stacks for intelligent Edge systems

Microchip Technology Inc. announces adding nine fresh technology- and



application-centric solution stacks to bolster its extensive mid-range FPGA (field programmable

gate array) and SoC (system-on-chip) support repertoire. Their precision customisation sets apart the PolarFire FPGA intelligent edge solution stacks. Unlike generic solutions that aim to cover a broad range of application domains, these stacks are meticulously fine-tuned to address specific technological and market requirements, encompassing intricate intellectual property, practical development kits, reference models, and more. The edge solutions and stacks for PolarFire FPGA and SoC platforms encompass various applications, including smart embedded vision with features like H.264 compression, HDMI, Serial digital interface, and CoaXpress, as

well as solutions for motor control and OPC/UA for the industrial edge.

Microchip Technology

<https://www.microchip.com>

Tensor Unit for enhancing AI performance and efficiency

Semidynamics has launched the RISC-V Tensor Unit designed for high-speed AI applications, leveraging its adaptable 64-bit cores. The Tensor Unit delivers great computational strength for AI applications, particularly focusing on the primary calculations of large language models (LLMs) that involve fully connected layers optimally executed as matrix multiplications. Built on Semidynamics' RVV1.0 Vector Processing Unit, it utilises existing vector registers for matrix storage, allowing it to efficiently handle layers requiring matrix multiplication, like Fully Connected and Convolution layers, while also managing activation function layers more effectively than independent neural processing units (NPU). Additionally, its integration with any RISC-V vector-supported Linux without needing modifications, due to its use of vector registers for data storage and avoidance of introducing new architecturally visible states, enhances its compatibility and functionality in AI tasks.

Semidynamics

<https://semidynamics.com/en>

Versatile Wi-Fi HaLow SoC for IoT applications

Newracom has launched the NRC7394: Wi-Fi HaLow system-on-chip (SoC). The SoC stands out in the Wi-Fi HaLow market, catering to a range of IoT



industries with its operation in the 750-950MHz spectrum. Its versatile capabilities allow for channel bandwidths of 1/2/4MHz, accommodating PHY rates from 150kbps to 15Mbps, suitable for diverse applications, from

low-rate sensors to high-rate surveillance cameras. The SoC features HSPI and UART host interfaces, alongside SPI, I2C, UART, PWM, auxiliary ADC, and GPIOs, with low-leakage retention memory for rapid wake-up from deep sleep mode. It includes a powerful integrated power amplifier for +17dBm TX output power and compatibility with external front-end module (FEM) devices for applications requiring higher transmission power.

Newracom

<https://newracom.com>

Ultra-rugged computers-on-modules for mission-critical applications

Congatec has launched six new computers-on-modules, the conga-TC675r COM Express Compact modules. The modules, powered by 13th Gen Intel Core processors and Intel's Raptor



Lake microarchitecture, are designed for harsh environments and extreme temperatures, ranging from -40°C to +85°C. They target

OEM applications in sectors like mining, construction, agriculture, and forestry, especially in rail and off-road vehicles. With soldered RAM, these modules meet standards for shock and vibration resistance, making them fit for stationary devices and outdoor applications in areas prone to natural disasters like earthquakes. They feature up to 14 cores and 20 threads, supported by LPDDR5x memory, for parallel processing and multitasking.

Congatec

<https://www.contec.com>

TEST & MEASUREMENT

True RMS benchtop digital multimeter

Triplett Test Equipment & Tools have launched the Triplett MM700B True root mean square (RMS) benchtop